Set 19: Tautology, contradiction and equivalence

Local outcome LOGIC.TRUTH.02: Classify compound statements as tautologies, contradictions or contingencies and test equivalence with a table

Formal logic set · genre: formal logic · MP7, MP3

MAIN: Classify statements as tautologies, contradictions or contingencies, and test equivalence
BACKWARD 1: Fill in truth tables
BACKWARD 2: Always, never, sometimes
FORWARD 1: ADVANCED APPLICATION: Equivalences by truth table
FORWARD 2: EXPERT CHALLENGE: Laws of logic

Which statement is a TAUTOLOGY (true in every row of its truth table)?

A. P or Q

B. (P and Q) or not P or not Q

C. P and Q and not Q

D. if P then Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q1
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which statement is a CONTRADICTION (false in every row)?

A. P or not P

B. P or Q

C. if P then Q

D. P and Q and not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q2
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Complete the truth table for "(if P then Q) or (if Q then P)".

Fill in T or F in every empty cell.

P	Q	if P then Q	if Q then P	(if P then Q) or (if Q then P)
T	T			
T	F			
F	T			
F	F			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q3
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"not (P and Q)" is logically equivalent to "(not P) or (not Q)".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q4
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

How is "if P then Q" classified?

A. a contingency: true in some rows and false in one

B. a contradiction

C. neither true nor false

D. a tautology

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q5
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Complete the truth table for "P or not P".

Fill in T or F in every empty cell.

P	not P	P or not P
T		
F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q1
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

Complete the truth table for "P and not P".

Fill in T or F in every empty cell.

P	not P	P and not P
T		
F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q2
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

Complete the truth table for "not P or Q".

Fill in T or F in every empty cell.

P	Q	not P or Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q3
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

How many rows does a truth table for P and Q have?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q4
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

A truth table for three statements P, Q and R has 6 rows.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q5
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

A statement is true in every row of its table. What is it called?

A. a contingency
B. a tautology
C. a premise
D. a contradiction

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q1
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

A statement is false in every row. What is it called?

A. a tautology
B. a contingency
C. a contradiction
D. a conclusion

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q2
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

A contingency is true in every row of its truth table.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q3
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

A column reads T, T, F, T. How is the statement classified?

A. it cannot be told
B. a contradiction
C. a tautology
D. a contingency

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q4
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"7 is odd and 7 is not odd" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q5
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Complete the table to compare "not (P or Q)" with "(not P) and (not Q)".

Fill in T or F in every empty cell.

P	Q	not (P or Q)	(not P) and (not Q)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q1
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"if P then Q" is equivalent to "if Q then P".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q2
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which statement is equivalent to "not (if P then Q)"?

A. not P or Q

B. P and not Q

C. not P and not Q

D. if not P then not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q3
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

How many of these are tautologies? "P or not P"; "P and not P"; "if P then Q"; "(if P then Q) or (if Q then P)"; "P iff not P".

Tautologies: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q4
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

The negation of a tautology is also a tautology.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q5
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which statement is equivalent to "P or (P and Q)"?

A. P
B. P and Q
C. P or Q
D. Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q1
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"((if P then Q) and P) implies Q", written "if ((if P then Q) and P) then Q", is a tautology.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q2
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"if ((if P then Q) and Q) then P" is a tautology.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q3
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Of the 16 possible truth-table columns for statements in P and Q, how many are contingencies?

Contingencies: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q4
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which pair are equivalent?

A. "not (P and Q)" and "not P and not Q"
B. "P and (Q or R)" and "(P and Q) or R"
C. "if (P or Q) then R" and "(if P then R) or (if Q then R)"
D. "if P then (Q and R)" and "(if P then Q) and (if P then R)"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q5
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Set 20: What is on top? Stacking and occlusion

Local outcome LOGIC.SPATIAL.03: Infer stacking order from what is visible and predict views of a stack

Puzzle set · genre: occlusion and stacking · MP7, MP1

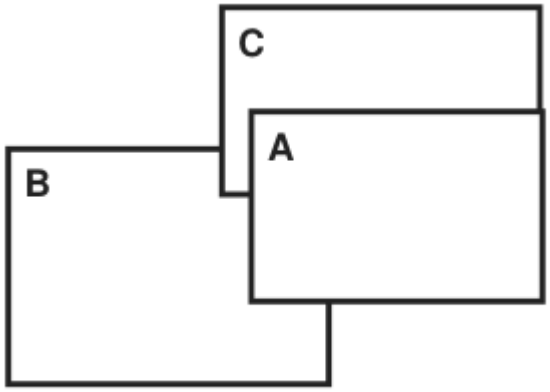
MAIN: Infer stacking order from what is visible

BACKWARD 1: Strategy: the cut-off sheet is underneath

FORWARD 1: ADVANCED APPLICATION: Larger stacks

FORWARD 2: EXPERT CHALLENGE: How much does a picture tell?

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 3 sheets: A, B, C. Put the sheets in order from the BOTTOM sheet to the TOP sheet.

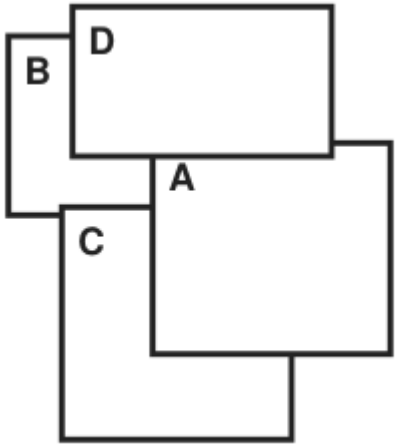


Put these in order:

A sheet A
B sheet B
C sheet C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q1
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. List the sheets starting with the one at the BOTTOM and ending with the one on TOP.



Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D

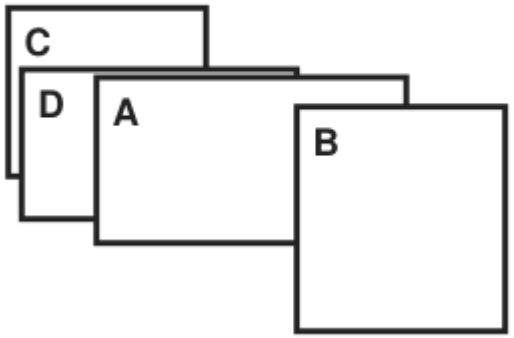
REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q2
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Four round discs are stacked with their centers lined up, from bottom to top: W (radius 5 cm), X (radius 3 cm), Y (radius 4 cm), Z (radius 1 cm). Looking straight down from above, which discs can be seen?

A. W and Z
B. Z only
C. W, Y and Z
D. W, X, Y and Z

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q3
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. Put the sheets in order from the BOTTOM sheet to the TOP sheet.



Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q4
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Square cards are placed one at a time on an 8 by 8 grid, each covering a block of cells (rows and columns numbered 1 to 8; "rows 1-4, columns 1-4" means a 4 by 4 block). In order: A covers rows 1-4, columns 1-4; B covers rows 3-6, columns 3-6; C covers rows 2-5, columns 2-5; D covers rows 1-3, columns 4-7; E covers rows 5-7, columns 1-3. Seen from above, how many of the five cards can be seen at least partly?

Cards seen: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q5
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Where sheets X and Y overlap, the outline of X runs straight through and the outline of Y stops at X's edge. What follows?

A. X is under Y.

B. It cannot be told.

C. Y is under X.

D. They are side by side.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q1
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Pairs found: P under Q, Q under R, P under R. Order the sheets from bottom to top.

Put these in order:

P sheet P

Q sheet Q

R sheet R

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q2
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

A sheet whose whole outline can be seen must be on top of every sheet it overlaps.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q3
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Discs are stacked with centers lined up, from bottom to top: radius 2 cm, then 6 cm. Which can be seen from above?

A. only the 2 cm disc
B. both discs
C. neither disc
D. only the 6 cm disc

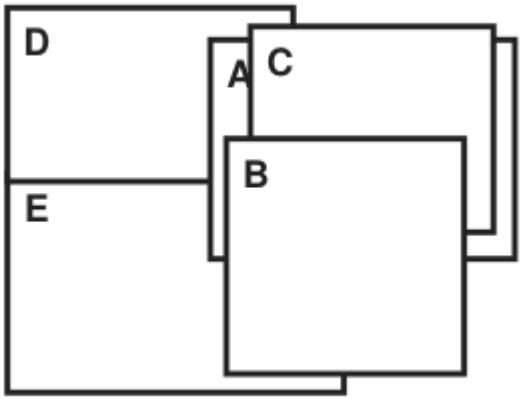
REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q4
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Five sheets are stacked. Exactly one of them is on top of all four others. How many sheets are under it?

Sheets under it: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q5
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. Put the five sheets in order from the BOTTOM sheet to the TOP sheet.

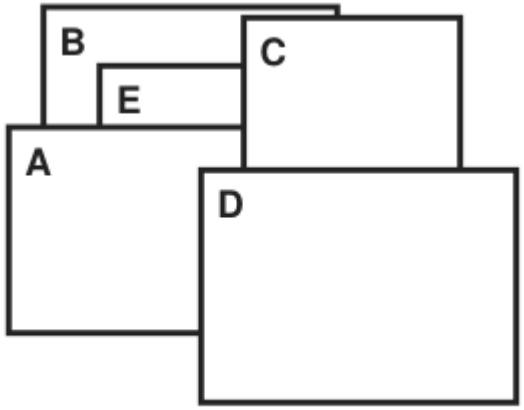


Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D
E sheet E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q1
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. List the five sheets starting with the BOTTOM sheet and ending with the TOP sheet.



Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D
E sheet E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q2
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Cards are placed one at a time on an 8 by 8 grid: A covers rows 2-3, columns 2-3; B covers rows 1-4, columns 1-4; C covers rows 4-6, columns 4-6; D covers rows 1-2, columns 5-6; E covers rows 3-6, columns 1-4. Seen from above, how many of the five cards can be seen at least partly?

Cards seen: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q3
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Four discs with centers lined up are stacked, bottom to top: radius 6, 2, 5, 3 (cm). How many can be seen from above?

A. 3
B. 1
C. 2
D. 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q4
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

If two sheets do not overlap at all, the picture from above cannot show which one is higher.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q5
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Three sheets: A and B do not overlap each other, and sheet C overlaps both and is on top of both. How many different bottom-to-top orders are consistent with the picture from above?

Orders: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q1
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Discs of radius 1, 2, 3 and 4 cm are stacked with centers lined up. In which stacking order (bottom to top) can ALL four discs be seen from above?

A. 3, 4, 2, 1
B. 4, 3, 2, 1
C. 1, 2, 3, 4
D. 4, 2, 3, 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q2
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Discs of radius 1, 2, 3 and 4 cm are stacked in some order with centers lined up. Over all 24 stacking orders, in how many is EXACTLY one disc visible from above?

Orders: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q3
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

From a picture of a stack of sheets, the TOP sheet can always be identified if every sheet overlaps at least one other sheet.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q4
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Cards on a grid: P covers rows 1-3, columns 1-3; Q covers rows 2-4, columns 2-4. P was placed first. Which cell shows card P?

A. row 1, column 2

B. row 3, column 3

C. row 2, column 2

D. row 4, column 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q5
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Set 21: Valid inference forms

Local outcome LOGIC.INFERENCE.01: Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Formal logic set · genre: formal logic · MP3

MAIN: Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism
BACKWARD 1: When is a conditional true?
BACKWARD 2: Using "or" and "not"
FORWARD 1: ADVANCED APPLICATION: Longer chains of valid steps
FORWARD 2: EXPERT CHALLENGE: Valid forms with several variables

Premises: "If the reservoir is full, the dam gates open." "The reservoir is full." What conclusion follows?

A. The reservoir is not full.

B. The dam gates do not open.

C. Nothing follows.

D. The dam gates open.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q1
Logic, Set 21: Valid inference forms · MAIN · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "If a number is divisible by 9, it is divisible by 3." "The number 52 is not divisible by 3." What follows?

A. 52 is divisible by 3.
B. 52 is divisible by 9.
C. 52 is not divisible by 9.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q2
Logic, Set 21: Valid inference forms · MAIN · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Arrange the steps of the argument so that each line follows from earlier ones and the conclusion comes last.

Put these in order:

A If the trail is icy, the park closes.
B If the park closes, the race is cancelled.
C The trail is icy.
D So the park closes.
E So the race is cancelled.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q3
Logic, Set 21: Valid inference forms · MAIN · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "The key is in the drawer or in the coat." "The key is not in the drawer." What follows?

A. The key is in neither place.
B. The key is in the coat.
C. The key is in the drawer.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q4
Logic, Set 21: Valid inference forms · MAIN · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

This argument is valid: "If P then Q. If Q then R. So if P then R."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q5
Logic, Set 21: Valid inference forms · MAIN · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"If P then Q" is false when P is false and Q is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q1
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is false when P is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q2
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is true and P is true. What is Q?

A. true
B. either
C. neither
D. false

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q3
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is true and Q is false. What is P?

A. neither
B. either
C. false
D. true

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q4
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is true and Q is true. P must be true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q5
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"P or Q" is true and P is true. Then Q must be false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q1
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

"P or Q" is true and P is true. Then Q is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q2
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

"The meeting is on Monday or Tuesday." It is not on Monday. When is it?

A. Monday
B. Tuesday
C. neither day
D. it cannot be told

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q3
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

Which is the negation of "the light is on"?

A. the light is dim
B. the lamp is broken
C. the light is bright
D. the light is not on

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q4
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

"P or Q" is true and both P and Q could be true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q5
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

Premises: "If it is a holiday, the bank is closed." "If the bank is closed, Mara cannot deposit." "Mara deposited." What follows?

A. It is a holiday.
B. It is not a holiday.
C. The bank is closed.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q1
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "P or Q. If P then R. If Q then R. So R."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q2
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "Ana or Ben will host." "If Ben hosts, the party is on Friday." "The party is not on Friday." Who hosts?

A. both
B. Ben
C. Ana
D. it cannot be told

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q3
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Arrange into a valid argument ending with the conclusion "The fountain is off."

Put these in order:

A If the fountain is on, the pump runs.

B The pump does not run.

C So the fountain is off.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q4
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "If P then Q. Not Q. So not P."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q5
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "If P then (Q and R). Not R. So not P."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q1
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "If P then (Q or R). Not Q. So R."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q2
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "If A then B." "If C then D." "A or C." Which conclusion follows?

A. B or D
B. B and D
C. not B
D. A and C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q3
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

An argument has premises "P or Q" and "if P then R" and conclusion "Q or R". In how many of the 8 rows of its truth table are both premises true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q4
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

The argument "P or Q. If P then R. So Q or R." is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q5
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Set 22: Tempting invalid inferences

Local outcome LOGIC.INFERENCE.02: Recognize affirming the consequent and denying the antecedent, and show why they fail

Formal logic set · genre: formal logic · MP3, MP6

- MAIN: Recognize affirming the consequent and denying the antecedent
- BACKWARD 1: Valid forms first
- BACKWARD 2: A counterexample shows invalidity
- FORWARD 1: ADVANCED APPLICATION: Spot invalid forms in disguise
- FORWARD 2: EXPERT CHALLENGE: Invalid or just unconvincing?

"If a phone is waterproof, it survives a fall into a pool. Leo's phone survived a fall into a pool. So Leo's phone is waterproof." What is wrong with this argument?

A. A premise is false.
B. It denies the antecedent.
C. Nothing: it is modus ponens.
D. It affirms the consequent: the phone may have survived for another reason.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q1
Logic, Set 22: Tempting invalid inferences · MAIN · Q1/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"If n is a multiple of 4, then n is even. 6 is not a multiple of 4. So 6 is not even." What is wrong?

A. It denies the antecedent: not P does not give not Q.
B. Nothing: it is modus tollens.
C. It affirms the consequent.
D. The conditional is false.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q2
Logic, Set 22: Tempting invalid inferences · MAIN · Q2/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

This form is valid: "If P then Q. Q. So P."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q3
Logic, Set 22: Tempting invalid inferences · MAIN · Q3/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Use whole numbers to show that "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." is invalid: give a value of n that makes both premises true and the conclusion false.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q4
Logic, Set 22: Tempting invalid inferences · MAIN · Q4/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Which argument is VALID?

A. "If it rains, the game is off. The game is off. So it is raining."
B. "If it rains, the game is off. It is not raining. So the game is not off."
C. "If it rains, the game is off. So if the game is off, it rains."
D. "If it rains, the game is off. The game is not off. So it is not raining."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q5
Logic, Set 22: Tempting invalid inferences · MAIN · Q5/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"If P then Q. P. So Q." What is this form called?

A. modus tollens
B. affirming the consequent
C. modus ponens
D. denying the antecedent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q1
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

"If P then Q. Not Q. So not P." What is this form called?

A. denying the antecedent
B. modus tollens
C. modus ponens
D. affirming the consequent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q2
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

Modus ponens is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q3
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

Modus tollens is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q4
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

In modus ponens, which part of the conditional is given as a fact?

A. the "if" part
B. neither part
C. the "then" part
D. the negation of the "if" part

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q5
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

An argument form is invalid if there is one case with all premises true and the conclusion false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q1
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

"If x > 5, then x > 2. x > 2. So x > 5." Which value of x shows it is invalid?

A. 1
B. 6
C. 10
D. 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q2
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

A valid argument can have true premises and a false conclusion.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q3
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

"If an animal is a robin, it has wings. A bat has wings. So a bat is a robin." Which fact shows the form is invalid?

A. Some birds cannot fly.

B. Bats fly at night.

C. A bat has wings but is not a robin.

D. Robins have wings.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q4
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

If one case with true premises has a true conclusion, the argument is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q5
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

"The alarm sounds only if there is smoke. There is smoke. So the alarm sounds." Is this valid?

A. Yes: it is modus ponens.
B. No: "A only if S" means "if A then S"; from S, A does not follow.
C. No: the conclusion is false.
D. Yes: smoke always sets off alarms.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q1
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"Every square is a rectangle. This shape is not a square. So it is not a rectangle." This argument is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q2
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Show that "If x > 10, then x > 3. x is not greater than 10. So x is not greater than 3." is invalid: give a number x that makes the premises true and the conclusion false.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q3
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Which argument has the SAME form as "If it is a cat, it is a mammal. It is a mammal. So it is a cat."?

A. "If n is 2, n is even. n is even. So n is 2."
B. "n is 2 or n is 3. n is not 3. So n is 2."
C. "If n is 2, n is even. n is 2. So n is even."
D. "If n is 2, n is even. n is odd. So n is not 2."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q4
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Denying the antecedent and affirming the consequent are both invalid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q5
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"If a number is divisible by 4, it is even. 12 is even. So 12 is divisible by 4." The conclusion is true, so the argument is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q1
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

For whole numbers n from 1 to 30, the argument "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." has true premises for every even n. For how many of these n is the conclusion false?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q2
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Which argument is INVALID, even though every premise and the conclusion are true?

A. "If 9 is even, then 9 is a whole number. 9 is even. So 9 is a whole number."
B. "9 is odd or 9 is even. 9 is not even. So 9 is odd."
C. "If 9 is odd, then 9 is a whole number. 9 is a whole number. So 9 is odd."
D. "If 9 is odd, then 9 is a whole number. 9 is odd. So 9 is a whole number."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q3
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Valid: "If P then Q. If not P then Q. So Q."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q4
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Valid: "If P then Q. If Q then P. Not P. So not Q."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q5
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

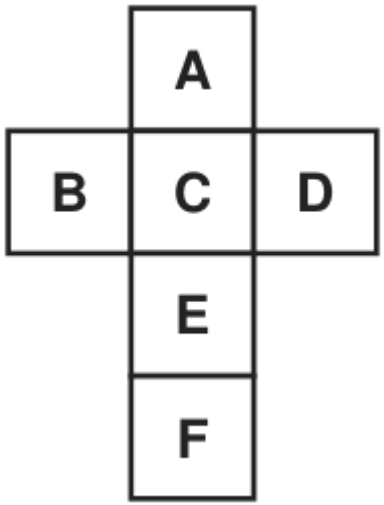
Set 23: Cube nets

Local outcome LOGIC.SPATIAL.04: Match a cube to its net: which faces are opposite and which are adjacent

Puzzle set · genre: cube nets · MP7, MP1

- MAIN: Match a cube to its net: opposite and adjacent faces
- BACKWARD 1: Strategy: skip one square to find the opposite face
- FORWARD 1: ADVANCED APPLICATION: Harder nets
- FORWARD 2: EXPERT CHALLENGE: Counting and designing nets

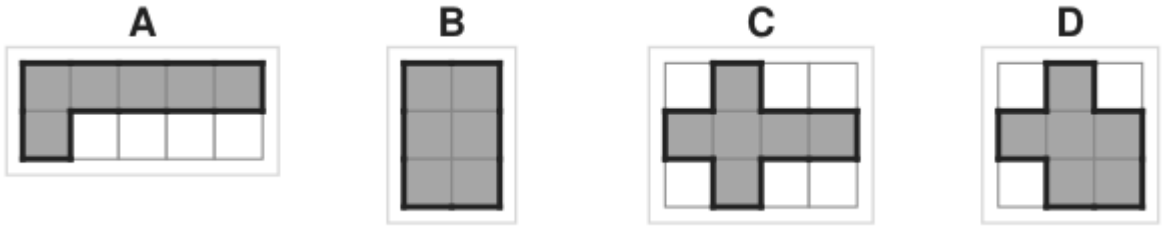
The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is on the face OPPOSITE the face A?



A. C
B. E
C. B
D. F

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q1
Logic, Set 23: Cube nets · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

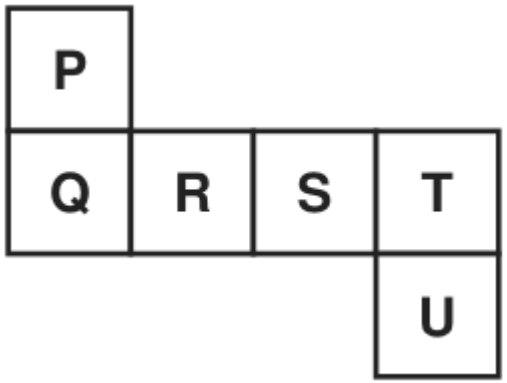
Each figure is made of six squares. Which figure can be folded along the lines between squares into a closed cube (with no overlapping squares)?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q2
Logic, Set 23: Cube nets · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face P, Q and R of this net, which letter is on the opposite face?

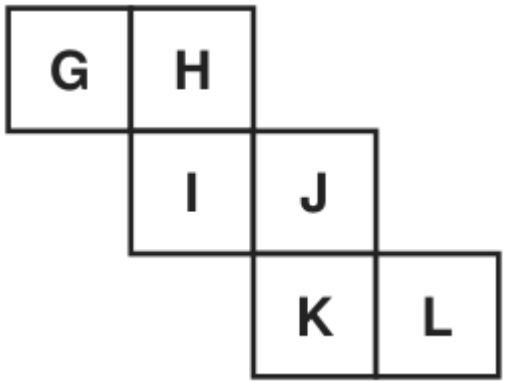


Choose one column for each row. Each column is used once.

	S	T	U
P			
Q			
R			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q3
Logic, Set 23: Cube nets · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

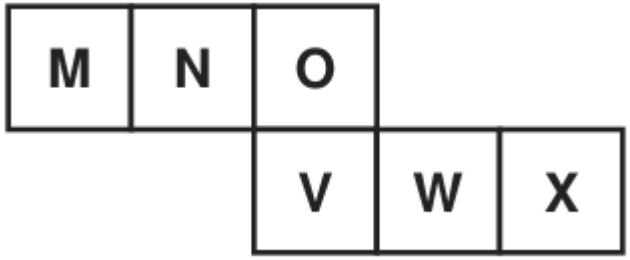
The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which two letters can NEVER be seen at the same time when you look at the cube?



A. G and H
B. H and J
C. H and I
D. G and J

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q4
Logic, Set 23: Cube nets · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. On the cube made from this net, faces M and W share an edge.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q5
Logic, Set 23: Cube nets · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

A straight row of four squares, numbered 1 to 4, is part of a cube net. Which square is opposite square 1?

A. 2
B. 3
C. 4
D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q1
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Two squares that share an edge in a net always share an edge on the cube.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q2
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

How many pairs of opposite faces does a cube have?

Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q3
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

On a cube, how many faces share an edge with any one face?

A. 4
B. 6
C. 3
D. 5

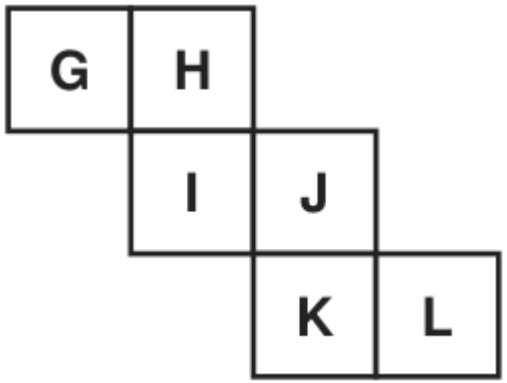
REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q4
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Two faces that are opposite on the cube can be seen at the same time.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q5
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

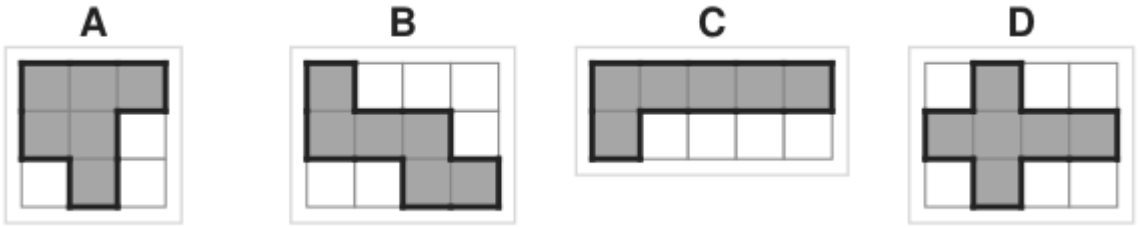
The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is opposite H?



A. G
B. J
C. I
D. K

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q1
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

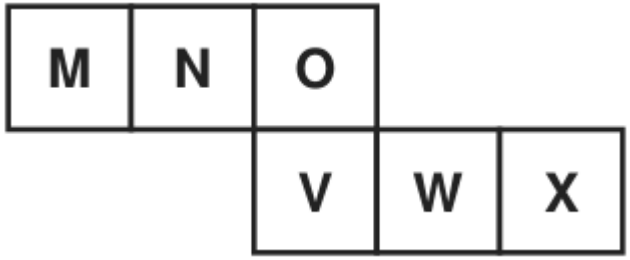
Each of the four figures is made of six squares. How many of them can be folded into a closed cube?



Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q2
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face M, N and V of this net, which letter is on the opposite face?



Choose one column for each row. Each column is used once.

	O	W	X
M			
N			
V			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q3
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

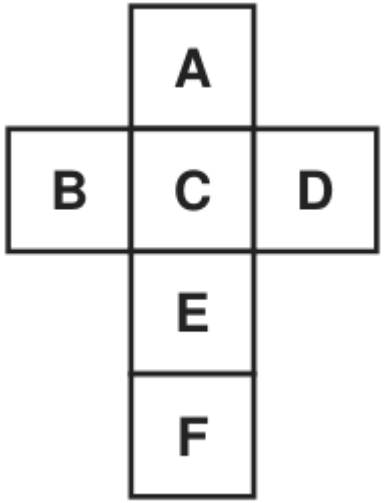
Every net of a cube has exactly 6 squares.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q4
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. In the cross-shaped net (A on top; B, C, D across; E and F below C), which two faces each share an edge with all of B, C and D?



A. B and D
B. E and F
C. A and E
D. A and F

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q5
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Count the different nets of a cube: shapes of six squares joined edge to edge that fold into a cube, where shapes that are turns or flips of each other count as the same. How many are there?

Nets: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q1
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The faces of a cube are colored so that faces sharing an edge always have different colors. What is the least number of colors needed?

Colors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q2
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

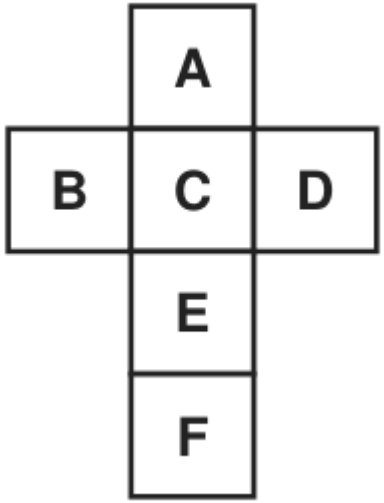
On a standard die, opposite faces add to 7. In a cross-shaped net with 1, 2, 3, 4 in the long column and 5, 6 on the two sides, the numbers can follow this rule.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q3
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. The numbers 1 to 6 are written on the cross-shaped net with A = 1, B = 2, C = 3, D = 5, E = 6 and F = 4. Which statement is true?



A. Every pair of opposite faces adds to 7.

B. Faces 3 and 4 touch.

C. Faces 2 and 6 are opposite.

D. Faces 1 and 2 are opposite.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q4
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

A cube net can contain four squares in a 2 by 2 block.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q5
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Set 24: Valid versus sound

Local outcome LOGIC.INFERENCE.03: Tell whether an argument is valid, and whether it is also sound

Formal logic set · genre: formal logic · MP3

MAIN: Tell whether an argument is valid, and whether it is also sound
BACKWARD 1: Is the form valid?
BACKWARD 2: Are the premises true?
FORWARD 1: ADVANCED APPLICATION: Classify harder arguments
FORWARD 2: EXPERT CHALLENGE: Validity, soundness and truth together

"All mammals breathe air. Whales are mammals. So whales breathe air." How is this argument best described?

A. sound but not valid
B. valid but not sound
C. invalid
D. valid and sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q1
Logic, Set 24: Valid versus sound · MAIN · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"All birds can swim. A sparrow is a bird. So a sparrow can swim." How is this argument best described?

A. sound but not valid
B. valid but not sound
C. valid and sound
D. invalid

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q2
Logic, Set 24: Valid versus sound · MAIN · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

An argument can be sound even if its conclusion is false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q3
Logic, Set 24: Valid versus sound · MAIN · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"If a number is divisible by 10, it is even. 14 is even. So 14 is divisible by 10." How is this described?

A. sound but not valid
B. valid but not sound
C. valid and sound
D. invalid, so it cannot be sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q4
Logic, Set 24: Valid versus sound · MAIN · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Which argument is SOUND?

A. "Every odd number is prime. 9 is odd. So 9 is prime."
B. "Every prime greater than 2 is odd. 13 is a prime greater than 2. So 13 is odd."
C. "If 13 is odd, 13 is prime. 13 is prime. So 13 is odd."
D. "Every prime is odd. 2 is prime. So 2 is odd."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q5
Logic, Set 24: Valid versus sound · MAIN · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"If P then Q. P. So Q." is a valid form.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q1
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"If P then Q. Q. So P." is a valid form.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q2
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

A valid argument guarantees what?

A. The argument is convincing.
B. The premises are true.
C. If the premises are true, the conclusion is true.
D. The conclusion is true.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q3
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"P or Q. P. So not Q." is a valid form.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q4
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

An argument with a true conclusion is always valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q5
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"Every rectangle has four equal sides" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q1
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every rectangle is a square" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q2
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every multiple of 2 is a multiple of 4" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q3
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every prime number is odd" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q4
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Water freezes at 10 degrees Celsius at normal air pressure" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q5
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every whole number that is divisible by 4 is divisible by 8. 12 is divisible by 4. So 12 is divisible by 8." How is it described?

A. valid but not sound
B. invalid
C. sound but not valid
D. valid and sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q1
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

A valid argument with a false conclusion can have all of its premises true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q2
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"No reptile has fur. A snake is a reptile. So a snake has no fur." How is it described?

A. sound but not valid
B. valid but not sound
C. invalid
D. valid and sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q3
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

If an argument is invalid, its conclusion must be false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q4
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Which statement is TRUE?

A. Every sound argument is valid.
B. Every argument with true premises is sound.
C. Every valid argument is sound.
D. Every argument with a true conclusion is sound.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q5
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Which combination is IMPOSSIBLE for an argument?

A. invalid, all premises true, conclusion true
B. invalid, all premises true, conclusion false
C. valid, all premises true, conclusion false
D. valid, a false premise, conclusion true

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q1
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

An argument whose premises can never all be true at the same time (they contradict each other) is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q2
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Of the 8 combinations (valid or invalid; premises all true or not; conclusion true or false), how many are impossible for an argument?

Impossible combinations: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q3
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"If x = 3, then x squared = 9. x squared = 9. So x = 3." For real x, this argument is:

A. sound: both premises can be true

B. invalid: x = -3 makes the premises true and the conclusion false

C. valid: squaring is unique

D. valid but not sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q4
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Adding a false premise to a sound argument keeps it valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q5
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Set 25: Moves and rearrangements

Local outcome LOGIC.SEQUENCE.01: Carry out stated moves on a row of tokens and find moves that reach a target

Puzzle set · genre: token moves · MP1, MP6

MAIN: Carry out stated moves on a row of tokens and find moves that reach a target
BACKWARD 1: Strategy: number the positions
FORWARD 1: ADVANCED APPLICATION: Fewest moves under different rules
FORWARD 2: EXPERT CHALLENGE: Moves that can never work

Tiles in a row each show a different letter. The row reads T, E, A, M. Move the leftmost tile to the far right end. Then swap the first two tiles. What is the row now, from left to right?

Put these in order:

T T
E E
A A
M M

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q1
Logic, Set 25: Moves and rearrangements · MAIN · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn D, C, B, A into A, B, C, D?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q2
Logic, Set 25: Moves and rearrangements · MAIN · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Which single move turns R, O, P, E into O, P, E, R?

A. Swap the two middle tiles.

B. Move E from the right end to the left end.

C. Swap R and O.

D. Move R from the left end to the right end.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q3
Logic, Set 25: Moves and rearrangements · MAIN · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn C, A, B, E, D into A, B, C, D, E?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q4
Logic, Set 25: Moves and rearrangements · MAIN · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Reversing the whole row twice puts every tile back where it started.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q5
Logic, Set 25: Moves and rearrangements · MAIN · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Row: K, I, T. Swap the first two tiles. What is the row?

Put these in order:

K K
I I
T T

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q1
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Row: P, A, N. Move the leftmost tile to the far right. What is the row?

Put these in order:

P P

A A

N N

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q2
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. How many pairs of tiles are in the wrong alphabetical order in C, B, A?

Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q3
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Swapping two neighboring tiles twice returns the row to how it was.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q4
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps neighboring tiles. Least number of moves from B, A, C to A, B, C?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q5
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn B, C, D, A into A, B, C, D?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q1
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn A, C, D, B into A, B, C, D?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q2
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Row: S, T, O, P. Move the tile in position 3 to the far left end, then swap the last two tiles. What is the row?

Put these in order:

S S
T T
O O
P P

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q3
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps neighboring tiles. What is the least number of moves that turn E, D, C, B, A into A, B, C, D, E?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q4
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. If one move swaps neighboring tiles, any arrangement of 4 tiles can be reached from any other.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q5
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move reverses any three tiles that are next to each other (for example A, B, C becomes C, B, A). Starting from A, B, C, D, E, can you reach B, A, C, D, E?

A. No: each move keeps every tile in an odd or an even position, and A would need to move from position 1 to position 2.

B. Yes: any arrangement can be reached with enough moves.

C. Yes: reverse A, B, C, then reverse B, A, ... in a few moves.

D. No: because each move changes three tiles at once.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q1
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move reverses any three neighboring tiles. Starting from A, B, C, D, E, how many different arrangements can be reached (counting the start)?

Arrangements: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q2
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps neighboring tiles. The row C, A, D, B, E must become A, B, C, D, E. Give the least number of moves m.

m = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q3
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps ANY two tiles (not only neighbors). D, C, B, A can be turned into A, B, C, D in 2 moves.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q4
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps ANY two tiles. What is the least number of moves that turn B, C, D, E, A into A, B, C, D, E?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q5
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Set 26: Quantifiers over a stated domain

Local outcome LOGIC.QUANT.01: Decide the truth of "all", "some" and "no" statements over a stated finite domain

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Decide the truth of "all", "some" and "no" statements over a stated domain

BACKWARD 1: Read "all", "some" and "no"

BACKWARD 2: Stay inside the stated domain

FORWARD 1: ADVANCED APPLICATION: Quantifiers with two conditions

FORWARD 2: EXPERT CHALLENGE: Nested quantifiers

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "All chess members have a bus pass."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q1
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member aged 16 has a bus pass."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q2
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is TRUE?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. All robotics members have a bus pass.

B. Every member is at least 16.

C. No art member has a bus pass.

D. Some chess member is 16.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q3
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). How many members make "the member is 15 or has no bus pass" true?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q4
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 10. Which statement is FALSE?

A. Some number in the domain is prime.
B. Every number in the domain is less than 10.
C. Some number in the domain is a multiple of 7.
D. No number in the domain is negative.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q5
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Some students walk to school" is true when ...

A. most students walk
B. at least one student walks to school
C. exactly one student walks
D. every student walks to school

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q1
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"No student walks to school" is true when ...

A. most students ride

B. some students do not walk

C. not even one student walks

D. exactly one student walks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q2
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All the lights are on" is true if exactly one light is off.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q3
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Some numbers in {2, 4, 6} are odd" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q4
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Every number in {3, 5, 7} is odd" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q5
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is {1, 2, 3}. "Every number in the domain is less than 4" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q1
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is {10, 20, 30}. "Some number in the domain is odd" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q2
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the days Monday to Friday. Which is true?

A. No day in the domain is a weekend day.
B. Every day in the domain starts with M.
C. Some day in the domain is Sunday.
D. Some day in the domain is Saturday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q3
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All prime numbers are odd" is true when the domain is {3, 5, 7, 11}.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q4
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All prime numbers are odd" is true when the domain is all whole numbers.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q5
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Every member without a bus pass is 16."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q1
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member with a bus pass is older than every robotics member."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q2
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 20. How many numbers make "the number is even and not a multiple of 3" true?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q3
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is FALSE?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. Every chess member is 15 or 17.

B. All members aged 17 play chess.

C. Some member has no bus pass.

D. No member aged 15 is in robotics.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q4
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is empty (it has no members). "Every member of the domain is green" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q5
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 5. "For every number x, there is a number y with y > x" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q1
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 5. "There is a number y such that for every number x, $y \geq x$" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q2
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is all whole numbers. Which statement is TRUE?

A. For every x and every y, $y > x$.

B. There is an x such that for every y, $y > x$.

C. For every x, there is a y with $y > x$.

D. There is a y such that for every x, $y > x$.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q3
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). For how many members x is the statement "some other member is the same age as x" true?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q4
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 9. Give a number x that makes "for every y in the domain, x + y > 5" true.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q5
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

Set 27: Negating quantified statements

Local outcome LOGIC.QUANT.02: Negate statements with "all", "some", "no", and nested quantifiers

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Negate statements with "all", "some", "no", and nested quantifiers
BACKWARD 1: The meaning of each quantifier
BACKWARD 2: Negating simple statements
FORWARD 1: ADVANCED APPLICATION: Negate quantified statements with conditions
FORWARD 2: EXPERT CHALLENGE: Nested quantifiers and their negations

What is the negation of "All the tickets were sold"?

A. At least one ticket was not sold.
B. All the tickets were unsold.
C. No ticket was sold.
D. Some tickets were sold.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q1
Logic, Set 27: Negating quantified statements · MAIN · Q1/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

What is the negation of "Some student brought a calculator"?

A. Some student did not bring a calculator.
B. Exactly one student brought a calculator.
C. Every student brought a calculator.
D. No student brought a calculator.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q2
Logic, Set 27: Negating quantified statements · MAIN · Q2/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

What is the negation of "No bridge in the town is closed"?

A. No bridge in the town is open.
B. Some bridge in the town is closed.
C. Every bridge in the town is closed.
D. Some bridge in the town is open.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q3
Logic, Set 27: Negating quantified statements · MAIN · Q3/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the whole numbers. The statement "Every whole number greater than 1 has an even square" is false. Give a number n that makes its negation true: a whole number greater than 1 whose square is odd.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q4
Logic, Set 27: Negating quantified statements · MAIN · Q4/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "Every square is a rectangle" is "No square is a rectangle".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q5
Logic, Set 27: Negating quantified statements · MAIN · Q5/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

"Some cards are red" is false if exactly one card is red.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q1
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All cards are red" is false if one card is blue.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q2
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"No card is red" is true if one card is red.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q3
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Not all cards are red" means ...

A. no card is red
B. every card is blue
C. exactly one card is red
D. at least one card is not red

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q4
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Some cards are red" and "some cards are not red" can both be true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q5
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

What is the negation of "the bus is late"?

A. the bus is not late
B. the bus is early
C. the train is late
D. the bus is very late

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q1
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

The negation of "$x \geq 4$" is "$x \leq 4$".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q2
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "the number is even and positive"?

A. the number is zero
B. the number is odd and negative
C. the number is odd or not positive
D. the number is even or positive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q3
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

"Not (the door is locked)" means the door is not locked.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q4
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "it rained on Monday or Tuesday"?

A. it did not rain on Monday
B. it rained on neither Monday nor Tuesday
C. it rained on both days
D. it rained on Wednesday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q5
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "Every student who studied passed"?

A. Some student who passed did not study.
B. Some student who studied did not pass.
C. No student who studied passed.
D. Every student who did not study failed.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q1
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the real numbers. What is the negation of "For every x, x squared ≥ 0"?

A. For every x, x squared ≤ 0.

B. There is an x with x squared ≥ 0.

C. There is an x with x squared < 0.

D. For every x, x squared < 0.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q2
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "Some prime number is even" is "Every prime number is odd" (for whole numbers greater than 1).

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q3
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The statement "Every whole number from 1 to 30 that ends in 9 is prime" is false. Give a number that shows it (a witness for its negation).

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q4
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "No dog is a cat" is "Every dog is a cat".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q5
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

What is the negation of "For every student, there is a club the student likes"?

A. There is a student who likes no club.

B. There is a club that no student likes.

C. No student likes any club.

D. For every student, there is a club the student does not like.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q1
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the whole numbers. What is the negation of "There is a number y such that for every x, y > x"?

A. There is no number y.
B. There is a y with $y \leq x$ for some x.
C. For every y and every x, $y \leq x$.
D. For every y, there is an x with $y \leq x$.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q2
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "For every x there is a y with $x + y = 0$" is "There is an x such that for every y, $x + y \neq 0$".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q3
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the whole numbers 1 to 10. The statement "For every x there is a y with y = 2x" is false in this domain. Give an x that shows it.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q4
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

For any statement about a domain, either the statement or its negation is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q5
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

Set 28: Routes and paths

Local outcome LOGIC.GRAPH.01: Trace routes in a drawn network and decide which routes exist under stated rules

Puzzle set · genre: routes and graphs · MP1, MP7

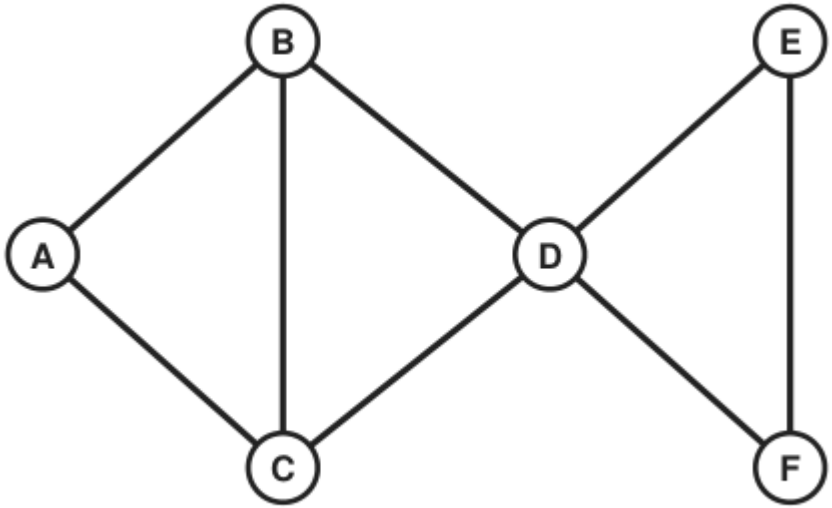
MAIN: Trace routes in a network and decide which routes exist

BACKWARD 1: Strategy: mark the route as you trace it

FORWARD 1: ADVANCED APPLICATION: Routes with conditions

FORWARD 2: EXPERT CHALLENGE: Every road, every town

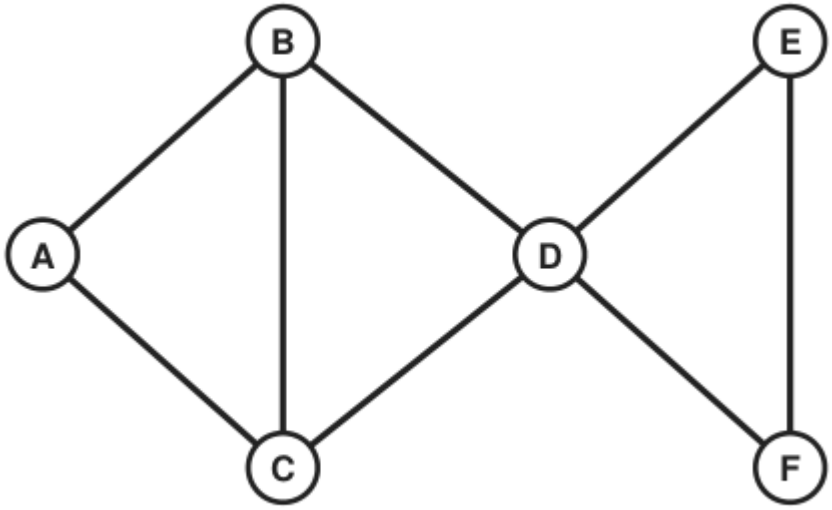
Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from A to F?



Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q1
Logic, Set 28: Routes and paths · MAIN · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

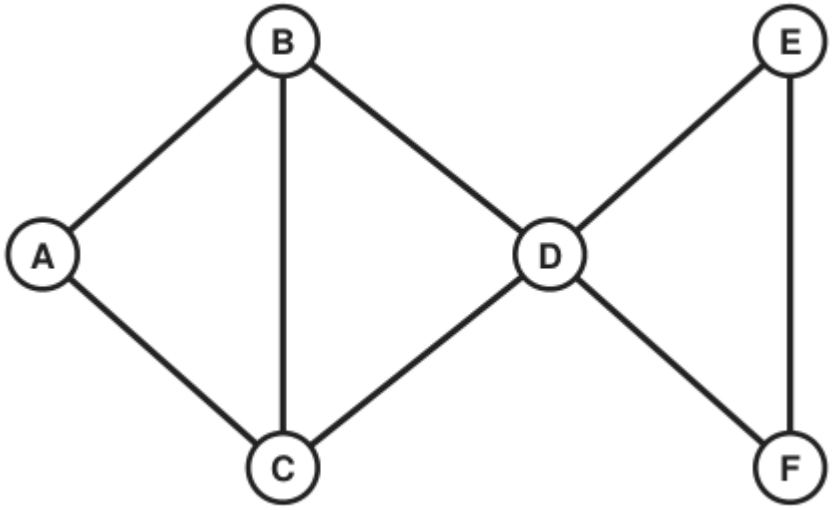
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many different routes go from A to F without visiting any town twice?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q2
Logic, Set 28: Routes and paths · MAIN · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

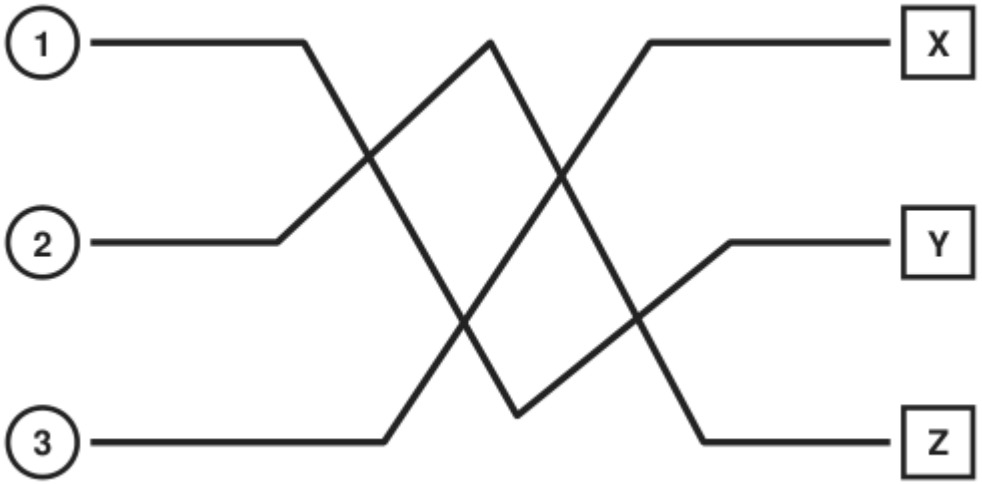
Roads are the straight lines; towns are the circles. Roads meet only at towns. Which town, if it were closed (with all its roads), would cut the network into two parts that cannot reach each other?



A. B
B. E
C. C
D. D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q3
Logic, Set 28: Routes and paths · MAIN · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

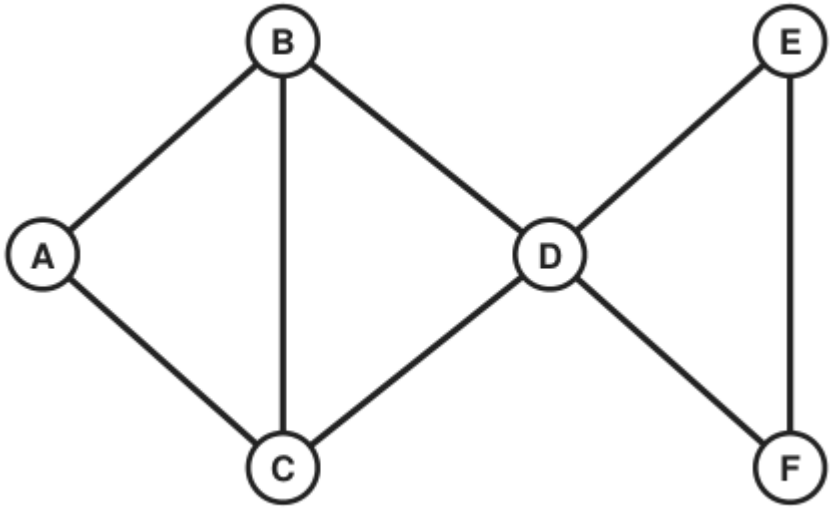
Walkers 1, 2 and 3 each follow their own line from the left to a gate on the right. Where lines cross, there is no turning: each walker keeps going straight along their own line. Which gate does walker 2 reach?



A. Z
B. Y
C. X
D. none of the gates

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q4
Logic, Set 28: Routes and paths · MAIN · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once (towns may be visited more than once).

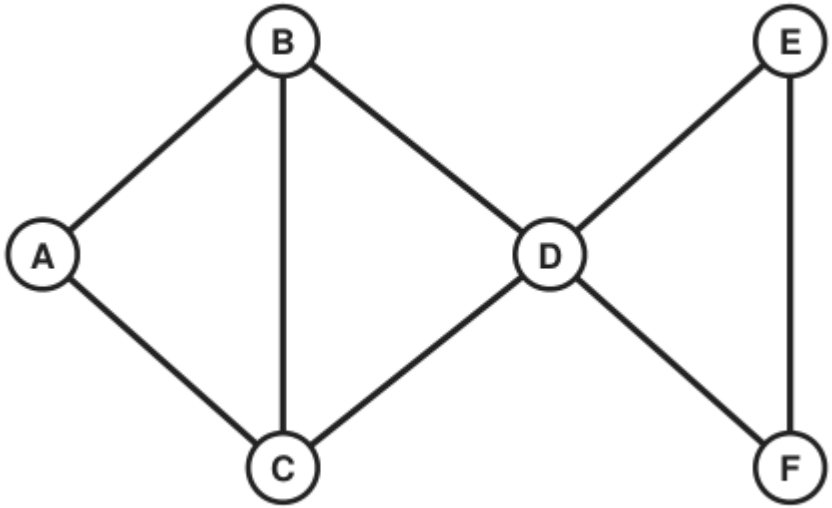


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q5
Logic, Set 28: Routes and paths · MAIN · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

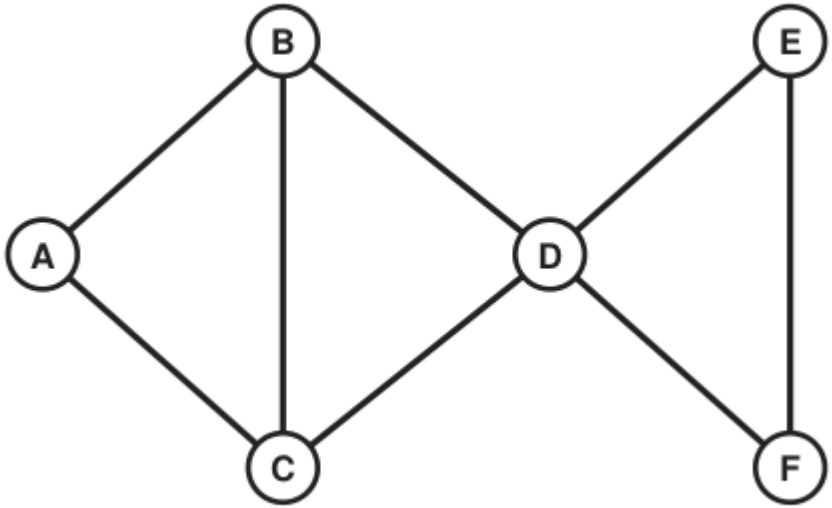
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many roads meet at town D?



Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q1
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

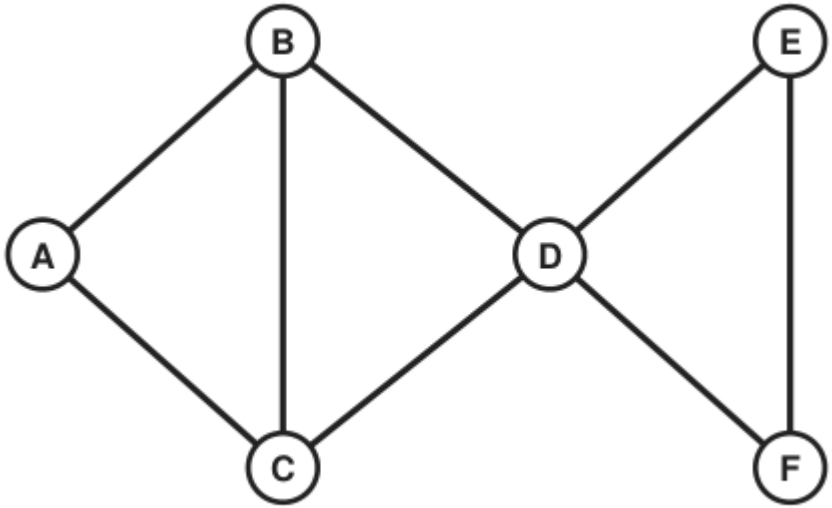
Roads are the straight lines; towns are the circles. Roads meet only at towns. Which towns can be reached from A using exactly one road?



A. only B
B. D
C. B and C
D. B, C and D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q2
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. There is a road directly from A to D.

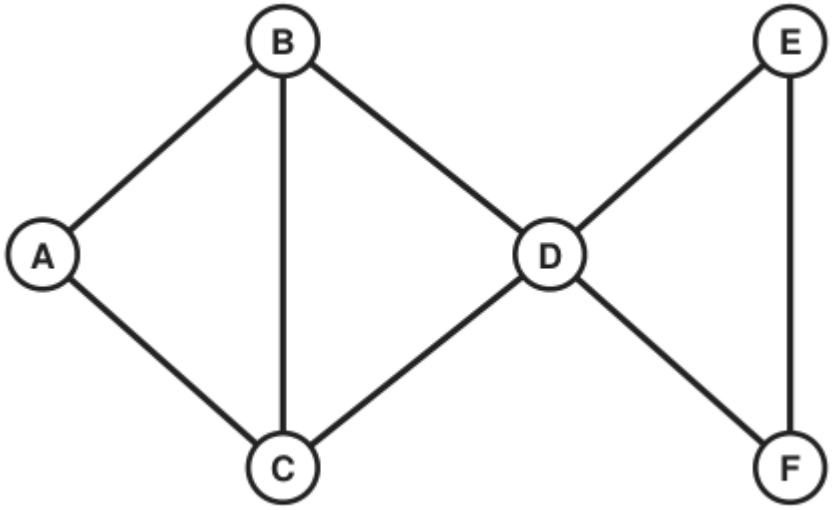


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q3
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from E to F without visiting a town twice?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q4
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

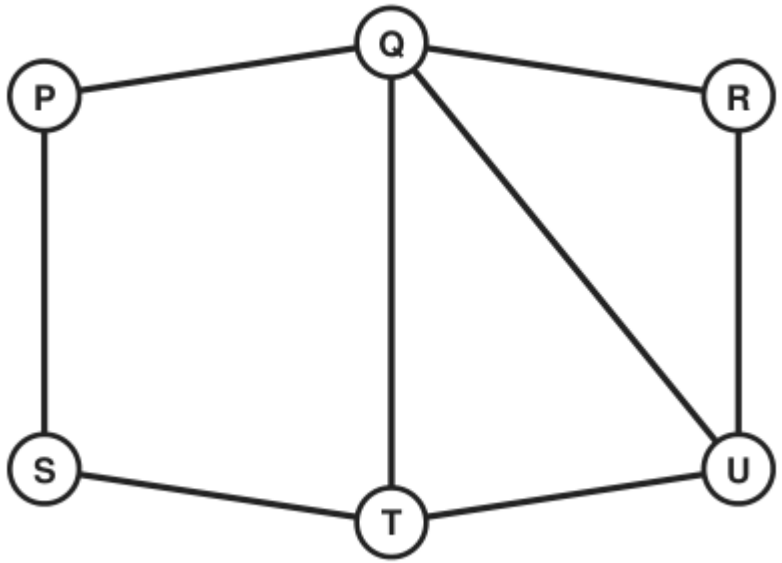
In a network, a town with 3 roads must be the start or end of a walk that uses every road once.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q5
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

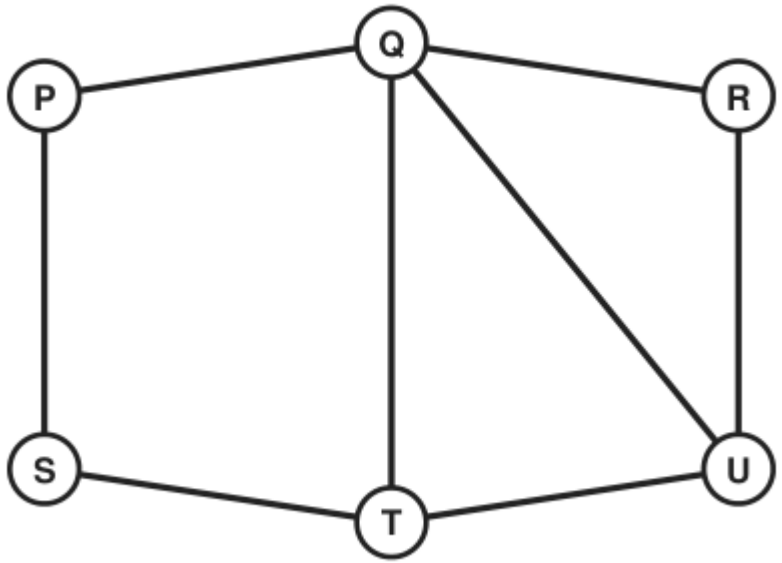
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from P to U without visiting a town twice?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q1
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

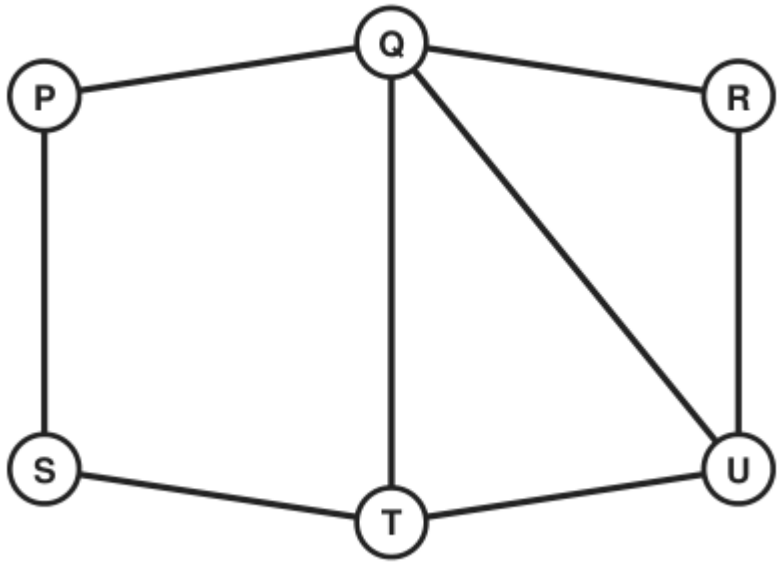
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from P to U (no town twice) pass through T?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q2
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once.

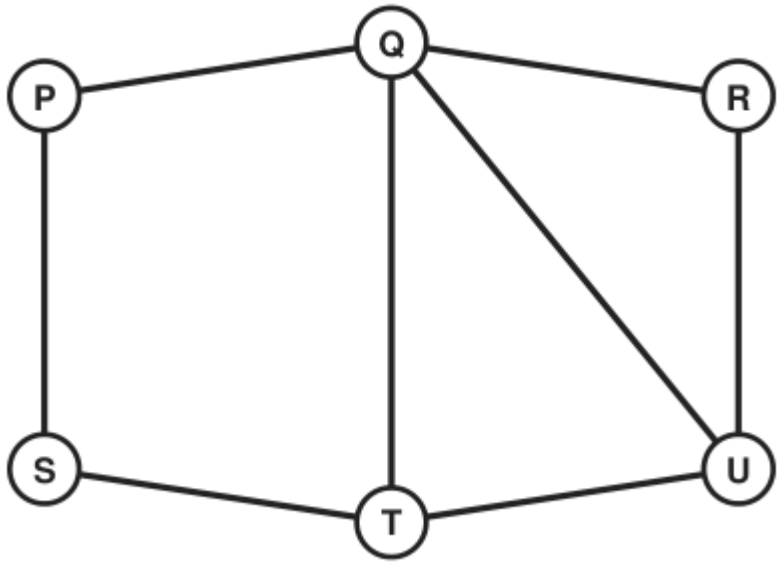


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q3
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

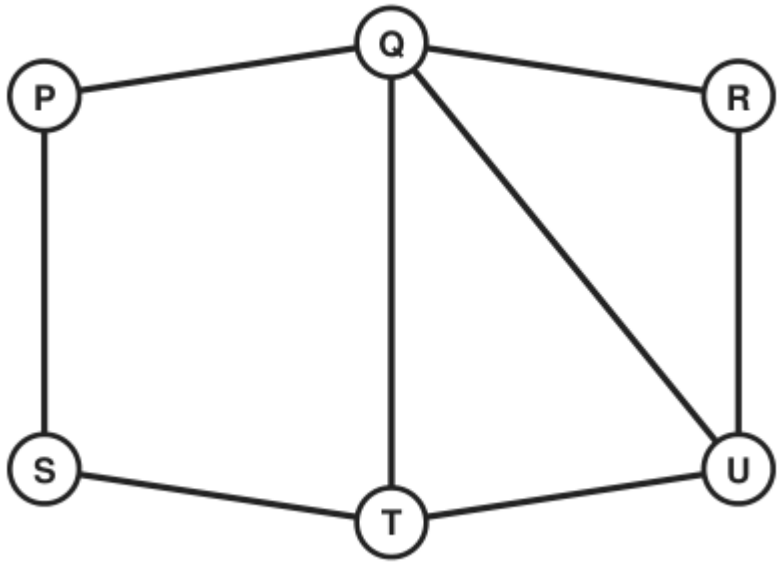
Roads are the straight lines; towns are the circles. Roads meet only at towns. Which single road, if closed, makes T unreachable from P by any route?



A. P-S
B. none of these roads
C. Q-T
D. S-T

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q4
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from S to R?



Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q5
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

In a network of 6 towns, the numbers of roads at the towns are 1, 2, 2, 3, 3 and 3. Every road has two ends, each at a town. How many roads are there?

Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q1
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

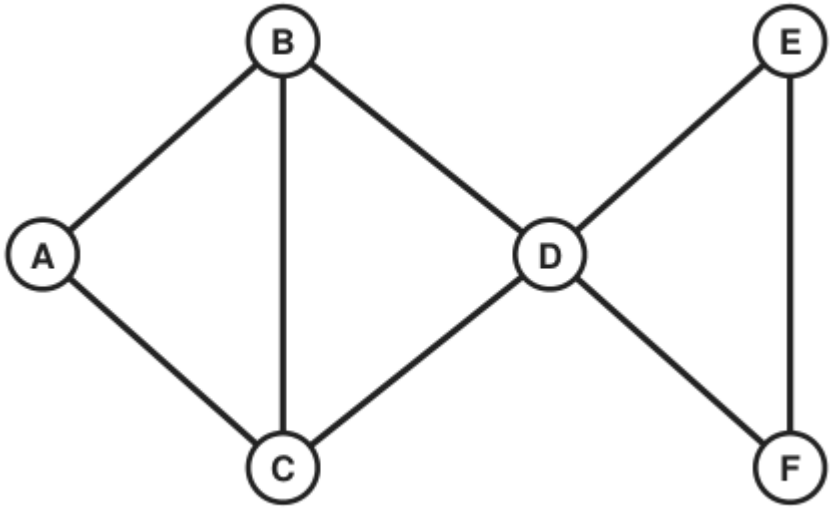
Some network has exactly three towns with an odd number of roads.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q2
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

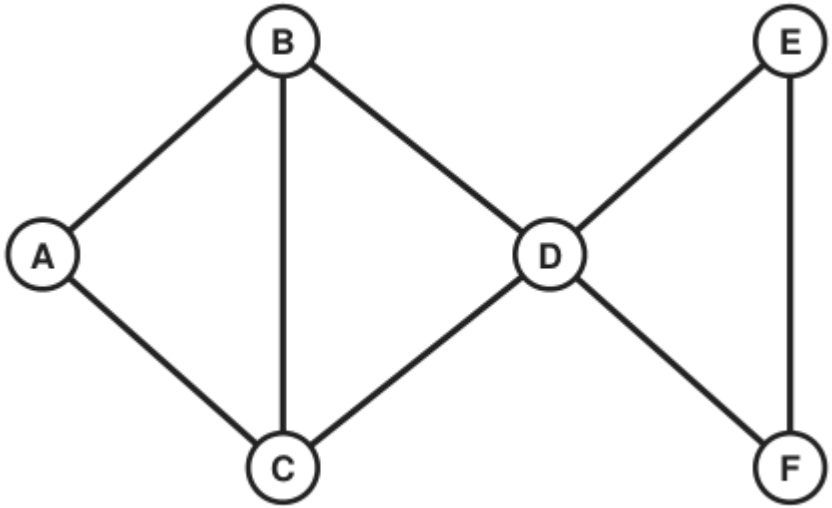
Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads that must be ADDED so that a walk can use every road exactly once and return to its starting town?



A. 2
B. 1
C. 3
D. 0

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q3
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from A to F visit EVERY town exactly once?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q4
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

A network has towns with these numbers of roads: 3, 3, 2, 2 and k. For a walk that uses every road exactly once to be possible in a connected network, at most two towns may have an odd number of roads. Give a whole number k from 1 to 6 that keeps this possible.

k = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q5
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Set 29: Witnesses and counterexamples

Local outcome LOGIC.QUANT.03: Give a witness for a "some" statement and a counterexample to an "all" statement

Formal logic set · genre: formal logic · MP3

- MAIN: Give a witness for a "some" statement and a counterexample to an "all" statement
- BACKWARD 1: Read "every" and "some" statements
- BACKWARD 2: Test a number against a property
- FORWARD 1: ADVANCED APPLICATION: Counterexamples in algebra
- FORWARD 2: EXPERT CHALLENGE: Witnesses that are hard to find

Give a witness for the true statement "Some whole number between 20 and 30 is prime."

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q1
Logic, Set 29: Witnesses and counterexamples · MAIN · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give a counterexample to "Every whole number that is divisible by 3 is odd."

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q2
Logic, Set 29: Witnesses and counterexamples · MAIN · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Which number is a counterexample to "Every number that ends in 1 is prime"?

A. 21
B. 31
C. 12
D. 11

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q3
Logic, Set 29: Witnesses and counterexamples · MAIN · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give real numbers a and b that are a counterexample to "For all real numbers a and b, (a + b) squared = a squared + b squared."

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q4
Logic, Set 29: Witnesses and counterexamples · MAIN · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

One example where a statement is true proves "Every number has this property."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q5
Logic, Set 29: Witnesses and counterexamples · MAIN · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

What kind of example disproves "Every bird can fly"?

A. a flying insect
B. a bird that can fly
C. a bird that cannot fly
D. an animal that is not a bird

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q1
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP3

What kind of example proves "Some square numbers are even"?

A. an odd square number

B. an even number that is not a square

C. every square number

D. an even square number, such as 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q2
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP3

"All cats are black" is shown true by one black cat.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q3
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP3

"All cats are black" is shown false by one white cat.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q4
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP3

"No number is negative" is shown false by the number 1.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q5
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP3

15 is prime.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q1
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

27 is divisible by 3 and odd.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q2
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

What is 7 squared minus 7?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q3
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

2 is both prime and even.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q4
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Which number is a multiple of 4 but not of 8?

A. 12
B. 10
C. 16
D. 8

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q5
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give a real number x that is a counterexample to "For every real number x, x squared ≥ x."

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q1
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give positive whole numbers a and b that are a counterexample to "If a divides b x c, then a divides b or a divides c", using c = 2.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q2
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Which value of n is a counterexample to "n squared + n + 1 is prime for every whole number n ≥ 1"?

A. 1

B. 4

C. 3

D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q3
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

The statement "For every real x, if x > 1 then x squared > x" has no counterexample.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q4
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give whole numbers a and b, each from 1 to 9, that show "a - b = b - a" can be false.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q5
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give a witness for "Some whole number from 1 to 100 leaves remainder 2 when divided by 3, remainder 3 when divided by 4, and remainder 4 when divided by 5."

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q1
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Is there a whole number n from 1 to 100 with n squared ending in the digit 2?

A. No: but only because 2 is even.

B. Yes: n = 12.

C. Yes: n = 22.

D. No: squares end only in 0, 1, 4, 5, 6 or 9.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q2
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give whole numbers a, b and c (each from 1 to 20) with a squared + b squared = c squared and a < b.

a = _____

b = _____

c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q3
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

How many counterexamples to "Every odd number from 1 to 30 is prime" are there in that range?

Counterexamples: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q4
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

To disprove "There is a whole number n with n + 5 = n", one example is enough.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q5
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Set 30: Sets and Venn diagrams

Local outcome LOGIC.SETS.01: Mark subset, intersection and complement relations on Venn diagrams

Formal logic set · genre: formal logic · MP7, MP6

MAIN: Mark subset, intersection and complement relations on Venn diagrams
BACKWARD 1: Regions of a two-circle diagram
BACKWARD 2: Set words: and, or, not
FORWARD 1: ADVANCED APPLICATION: Three-circle diagrams and counts
FORWARD 2: EXPERT CHALLENGE: Inclusion and exclusion

Circle A is the set of students in soccer and circle B the set of students in band. Mark the region for students in soccer but NOT in band.

Mark the regions on the Venn diagram of soccer (A), band (B) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q1
Logic, Set 30: Sets and Venn diagrams · MAIN · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Circle A is the set of even numbers and circle B the set of multiples of 3 (whole numbers 1 to 30 make up the outside box). Mark the region that contains 12.

Mark the regions on the Venn diagram of even (A), multiples of 3 (B) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q2
Logic, Set 30: Sets and Venn diagrams · MAIN · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Three circles: A (likes math), B (likes art), C (likes music). Mark every region for people who like EXACTLY two of the three subjects.

Mark the regions on the Venn diagram of math (A), art (B), music (C) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q3
Logic, Set 30: Sets and Venn diagrams · MAIN · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

In a class of 30 students, 18 play soccer, 12 play in the band, and 5 do both. How many students do neither?

Students: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q4
Logic, Set 30: Sets and Venn diagrams · MAIN · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Every member of set A is also in set B (A is a subset of B). On a two-circle diagram, mark the region that must be EMPTY.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q5
Logic, Set 30: Sets and Venn diagrams · MAIN · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark the region for things in BOTH A and B.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q1
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark the region for things in NEITHER A nor B.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q2
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark every region for things in A (whether or not they are in B).

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q3
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

How many regions does a two-circle Venn diagram have, counting the outside?

Regions: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q4
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark every region for things in B but not in A.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q5
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

The intersection of A and B contains things that are ...

A. in neither
B. in A or in B
C. in A and in B
D. in A but not B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q1
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

The union of A and B contains things that are ...

A. in A or in B (or both)
B. in exactly one
C. in neither
D. in A and in B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q2
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

The complement of A contains everything in the box that is not in A.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q3
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the union of A and B?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q4
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the intersection?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q5
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

Three circles A, B, C. Mark every region for things in A or B but NOT in C.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q1
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Three circles A, B, C. Mark every region for things in at least two of the sets.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q2
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Of 40 students, 22 take French, 17 take Spanish and 9 take both. How many take exactly one of the two?

Students: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q3
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Every member of A is in B, and no member of B is in C. On a three-circle diagram (A, B, C), mark every region that must be EMPTY.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q4
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

If A is a subset of B, then the intersection of A and B is A.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q5
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

In a group of 50 people, 25 like tea, 30 like coffee and 15 like both. How many like neither?

People: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q1
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Sets A, B, C have 10, 12 and 8 members. A and B share 4, A and C share 3, B and C share 2, and all three share 1. How many members are in the union?

Members: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q2
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

In a club of 20, s members swim and r members run; every member does at least one. Give possible numbers s and r if exactly 5 do both.

swim s = _____

run r = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q3
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Two sets can have 8 and 6 members and a union of 15 members.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q4
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Sets A and B are inside a box of 30 things. A has 12 things, B has 9, and A and B share 4. How many things are in the box but in neither A nor B?

Things: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q5
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

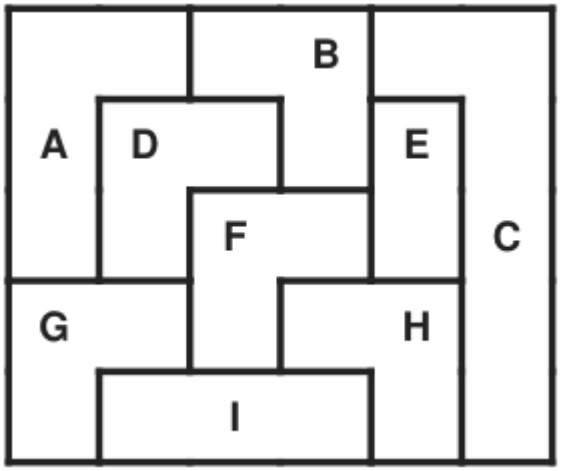
Set 31: Maps and neighbors

Local outcome LOGIC.GRAPH.02: Turn a map into a neighbor (adjacency) graph and reason about neighbor counts and colorings

Puzzle set · genre: region adjacency · MP7, MP1

MAIN: Turn a map into a neighbor graph and reason about neighbors and colorings
BACKWARD 1: Strategy: draw the neighbor graph
FORWARD 1: ADVANCED APPLICATION: Neighbor counts as clues
FORWARD 2: EXPERT CHALLENGE: Colorings: count and prove

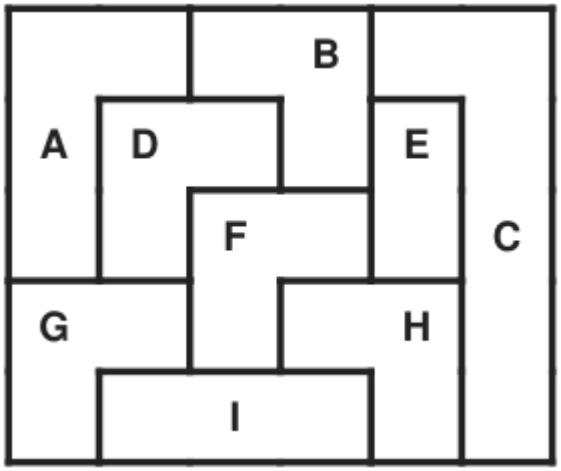
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region F have?



Neighbors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q1
Logic, Set 31: Maps and neighbors · MAIN · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

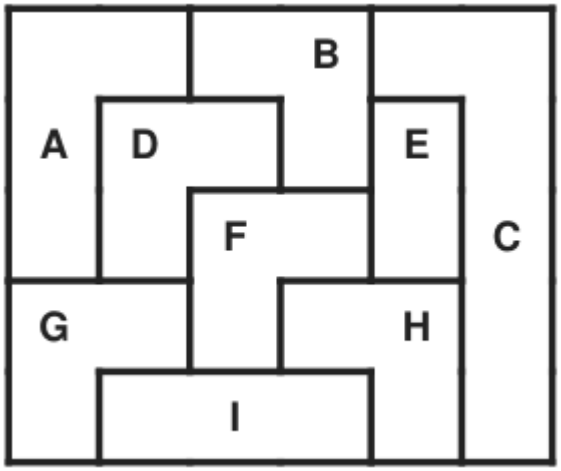
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which region has the most neighbors?



A. C
B. F
C. D
D. B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q2
Logic, Set 31: Maps and neighbors · MAIN · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions A and I are neighbors.

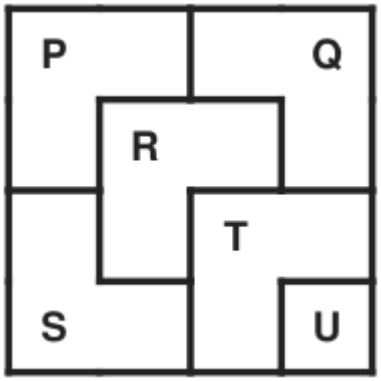


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q3
Logic, Set 31: Maps and neighbors · MAIN · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

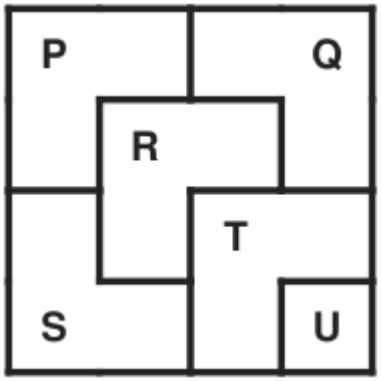
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Each region gets one color and neighbors must get different colors. What is the least number of colors needed for this map?



Colors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q4
Logic, Set 31: Maps and neighbors · MAIN · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Color each region red, blue or green so that neighbors differ. P is red and Q is blue. Give every region's color.

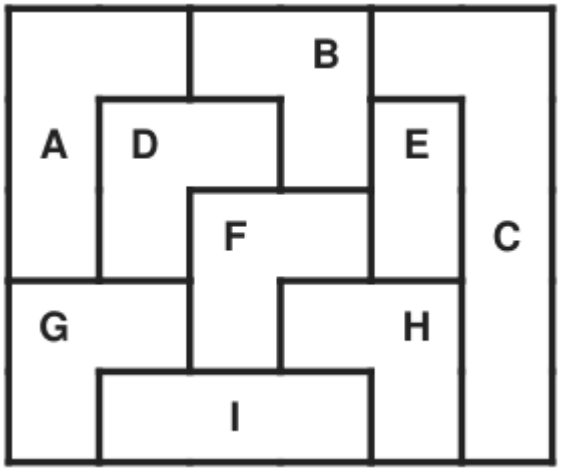


Choose one column for each row.

	red	blue	green
P			
Q			
R			
S			
T			
U			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q5
Logic, Set 31: Maps and neighbors · MAIN · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

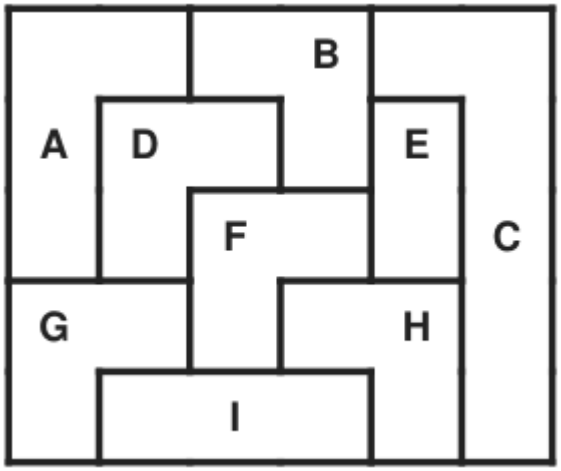
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region C have?



Neighbors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q1
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions B and D are neighbors.



A. True

B. False

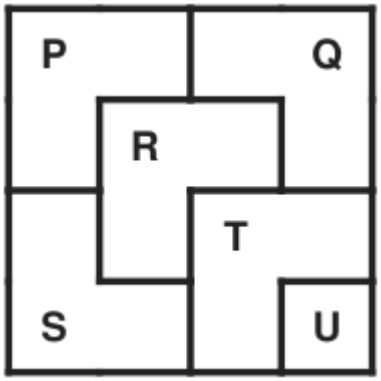
REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q2
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Two regions touch only at a single corner point. Are they neighbors under this rule?

A. Yes
B. It depends on their size
C. No
D. Only if they are the same color

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q3
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region U have?



Neighbors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q4
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

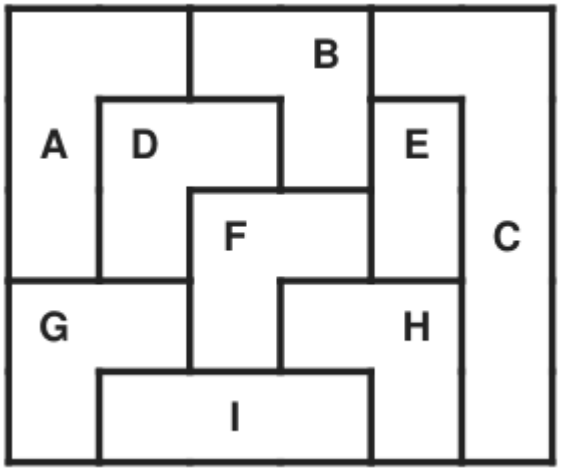
If three regions all border each other, a coloring with different colors for neighbors needs at least 3 colors.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q5
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

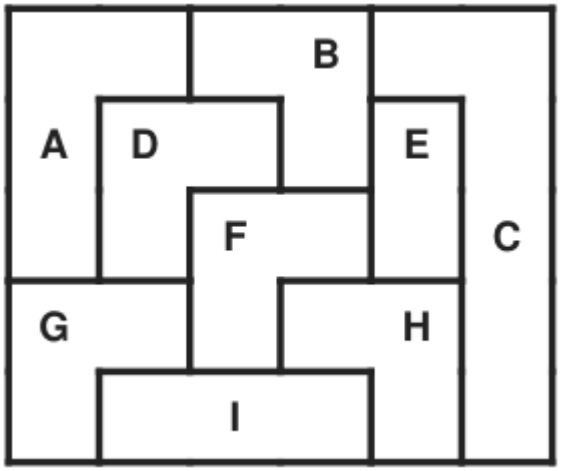
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the total of the neighbor counts of all nine regions?



Total: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q1
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

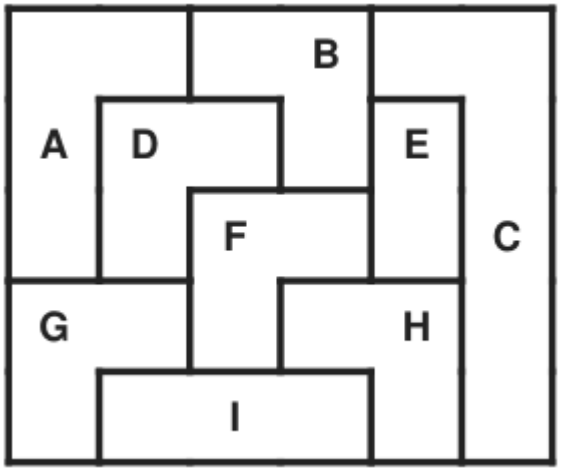
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many pairs of regions share a border?



Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q2
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

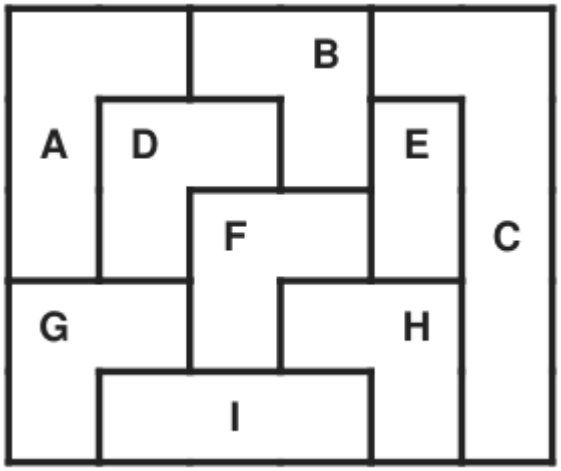
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the least number of colors needed so that neighbors differ?



Colors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q3
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. In any coloring of this map with 3 colors where neighbors differ, regions A and F have the same color.

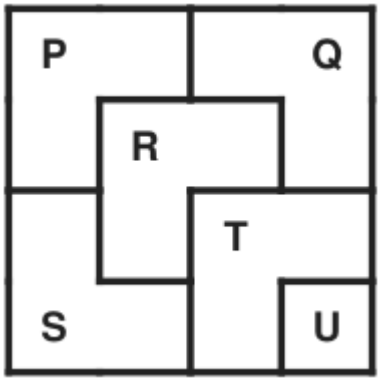


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q4
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

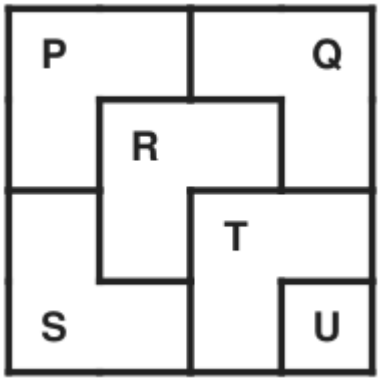
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which ONE region could be removed so that only 2 colors are needed for the rest?



A. S
B. U
C. T
D. R

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q5
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can the map be colored with red, blue and green so that neighbors differ?



Colorings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q1
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Some map drawn on a page needs at least 4 colors so that neighbors differ.

A. True

B. False

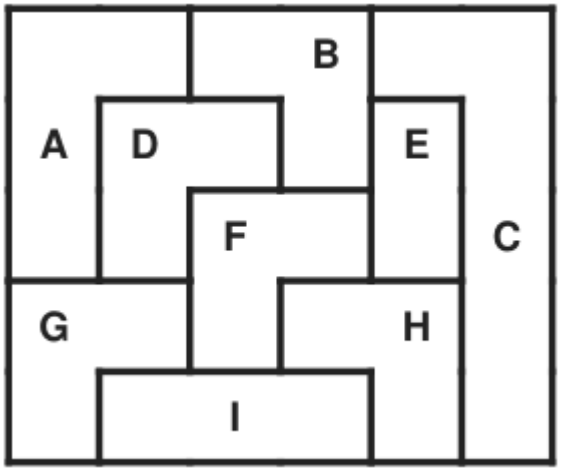
REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q2
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

A map has regions in a ring: region 1 borders 2, 2 borders 3, and so on, and the last region borders region 1 (and there are no other borders). Give a number of regions n, from 3 to 10, for which 2 colors are NOT enough.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q3
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can this map be colored with red, blue and green so that neighbors differ, if A is red?



Colorings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q4
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

If a map can be colored with 4 colors so that neighbors differ, it can also be colored with 3 colors.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q5
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Set 32: Categorical arguments

Local outcome LOGIC.SETS.02: Test categorical arguments with Venn diagrams, including the existence assumption

Formal logic set · genre: formal logic · MP3, MP7

- MAIN: Test categorical arguments with Venn diagrams, including the existence assumption
- BACKWARD 1: Mark statements on a Venn diagram
- BACKWARD 2: Valid means no counterexample
- FORWARD 1: ADVANCED APPLICATION: Harder categorical forms
- FORWARD 2: EXPERT CHALLENGE: Existence and hidden assumptions

Premise: "No reptile is warm-blooded." Circle A is reptiles and circle B is warm-blooded animals. Mark the region this premise says is EMPTY.

Mark the regions on the Venn diagram of reptiles (A), warm-blooded (B) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q1
Logic, Set 32: Categorical arguments · MAIN · Q1/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All violinists are musicians. All musicians practice. So all violinists practice." Is this valid?

A. No: some violinists may not practice.

B. Yes, but only if every musician is a violinist.

C. Yes: every violinist is inside the musicians, who are all inside those who practice.

D. No: the conclusion reverses a premise.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q2
Logic, Set 32: Categorical arguments · MAIN · Q2/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All squares are rectangles. Some rectangles are not rhombuses. So some squares are not rhombuses." Is this valid?

A. No: the rectangles that are not rhombuses may all lie outside the squares.

B. Yes: the conclusion is true.

C. Yes: squares are rectangles, so they share every property.

D. No: no rectangle is a square.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q3
Logic, Set 32: Categorical arguments · MAIN · Q3/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. Some C are A. So some C are B." This form is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q4
Logic, Set 32: Categorical arguments · MAIN · Q4/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All dragons breathe fire. So some dragons breathe fire." When does this follow?

A. Always: "all" includes "some".
B. Only if dragons are real animals.
C. Never: the premise is about all dragons.
D. Only if at least one dragon exists (the existence assumption).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q5
Logic, Set 32: Categorical arguments · MAIN · Q5/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B." Mark the region that must be EMPTY.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q1
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"No A is B." Mark the region that must be EMPTY.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q2
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"Some A is B." Mark the region where something must be.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q3
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"Some A is not B." Mark the region where something must be.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q4
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"All A are B" and "No A is B" can both be true if A has no members.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q5
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

An argument is valid if a situation exists with true premises and a false conclusion.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q1
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"All cats are mammals. All dogs are mammals. So all cats are dogs." Which situation shows it is invalid?

A. a world where all mammals are dogs

B. the real world: cats and dogs are both mammals but cats are not dogs

C. a world with no mammals

D. a world with no cats

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q2
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"All A are B. All B are C. So all A are C." is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q3
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"Some A are B. Some B are C. So some A are C." is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q4
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

Which premise pair makes "So no A is C" follow?

A. "All A are B." and "Some B are not C."

B. "Some A are B." and "No B is C."

C. "All A are B." and "No B is C."

D. "No A is B." and "No B is C."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q5
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"No A is B. All C are B. So no C is A." This form is valid.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q1
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. No C is A. So no C is B." This form is valid.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q2
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"Some athletes are vegetarians. All vegetarians eat beans. So ..." Which conclusion follows?

A. all athletes eat beans
B. no athlete eats meat
C. all bean eaters are athletes
D. some athletes eat beans

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q3
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

Premises: "All A are B" and "No B is C". On a three-circle diagram (A, B, C), mark every region that the premises make EMPTY.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q4
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. Some A are C. So some B are C." This form is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q5
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. All A are C. So some B are C." This form is valid without any existence assumption.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q1
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

Which added premise makes "All A are B. All A are C. So some B are C." valid?

A. At least one A exists.
B. All B are A.
C. At least one B exists.
D. At least one C exists.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q2
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

How many of these forms are valid (with no existence assumption)? (1) All A are B; all B are C; so all A are C. (2) No A is B; all C are A; so no C is B. (3) All A are B; some C are not B; so some C are not A. (4) Some A are B; all B are C; so some A are C.

Valid forms: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q3
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All unicorns have one horn" is true if no unicorns exist.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q4
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"No square is a circle. Some circles are shaded. So some shaded things are not squares." Is it valid?

A. No: shaded things may all be squares.
B. Yes: a shaded circle is not a square.
C. No: it needs squares to exist.
D. Yes, but only if all circles are shaded.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q5
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

Set 33: Word ladders

Local outcome LOGIC.WORD.02: Find shortest one-letter-change paths through a supplied word list

Puzzle set · genre: word ladders · MP1, MP7

MAIN: Find shortest one-letter-change paths through a supplied word list
BACKWARD 1: Strategy: change the letters that must change, one at a time
FORWARD 1: ADVANCED APPLICATION: Longer ladders and detours
FORWARD 2: EXPERT CHALLENGE: Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Put these four words in ladder order, starting with SAND, so that each word differs from the next by one letter.

Put these in order:

A SAND
B SEND
C SENT
D RENT

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q1
Logic, Set 33: Word ladders · MAIN · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps in a ladder from BAND to BOAT?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q2
Logic, Set 33: Word ladders · MAIN · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word fills the gap in the ladder SENT, ?, RANT?

A. BENT
B. RENT
C. PANT
D. SEND

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q3
Logic, Set 33: Word ladders · MAIN · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is one step from both BEND and BEAT.

word = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q4
Logic, Set 33: Word ladders · MAIN · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. A ladder from LAND to BELT can be done in 3 steps.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q5
Logic, Set 33: Word ladders · MAIN · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word is one step from BEST?

A. REST
B. BOAT
C. POST
D. RENT

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q1
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

In how many positions do the words MOST and PART differ?

Positions: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q2
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

A ladder from MAST to LAST can be done in one step.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q3
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from PORT to MOST?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q4
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

If two words differ in 3 positions, every ladder between them has at least 3 steps.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q5
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from BEND to PORT?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q1
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many different shortest ladders go from BAND to BOAT?

Ladders: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q2
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word needs the MOST steps to reach from SAND (using shortest ladders)?

A. BOLT
B. PORT
C. MOST
D. LAST

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q3
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is exactly one step from LAND and also one step from SEND.

word = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q4
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. BOAT is one step from BOLT.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q5
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from SAND to MOST?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q1
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Suppose PANT is removed from the list. What is the least number of steps from RANT to PART now? Enter 0 if no ladder is possible.

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q2
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Every ladder from BAND to BOAT passes through BEND.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q3
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many words on the list are exactly one step from BENT?

Words: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q4
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

A ladder must go from a 4-letter word to another 4-letter word that differs in all 4 positions. What is the least possible number of steps?

A. 3
B. 8
C. 4
D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q5
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Set 34: Translate clues into conditions

Local outcome LOGIC.CONSTRAINT.02: Translate puzzle clues into exact conditions and test candidate arrangements against them

Puzzle set · genre: clue translation · MP2, MP6

MAIN: Translate puzzle clues into exact conditions and test candidate arrangements against them
BACKWARD 1: Strategy: write each clue as a condition on numbers
FORWARD 1: ADVANCED APPLICATION: Harder clue words
FORWARD 2: EXPERT CHALLENGE: Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition says exactly the same thing as the clue "Exactly one locker lies between Cal and Dee"?

A. C - D = 2
B. |C - D| = 1
C. |C - D| >= 2
D. |C - D| = 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q1
Logic, Set 34: Translate clues into conditions · MAIN · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. Put the friends in locker order, from locker 1 to locker 5.

Put these in order:

A Ava
B Ben
C Cal
D Dee
E Eli

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q2
Logic, Set 34: Translate clues into conditions · MAIN · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. The arrangement Ava, Cal, Eli, Dee, Ben (lockers 1 to 5) breaks exactly one of the five clues.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q3
Logic, Set 34: Translate clues into conditions · MAIN · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. The arrangement Ava, Dee, Eli, Ben, Cal (lockers 1 to 5) breaks one clue. Which one?

A. Clue 2
B. Clue 5
C. Clue 1
D. Clue 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q4
Logic, Set 34: Translate clues into conditions · MAIN · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

A row has nine lockers numbered 1 to 9. Eli's locker must meet every condition: it is odd-numbered; it is not at either end (not 1 and not 9); and it is at least 2 lockers away from locker 5 (|E - 5| >= 2). Give one locker number Eli could have.

E = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q5
Logic, Set 34: Translate clues into conditions · MAIN · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava is somewhere left of Ben"?

A. A = B - 1

B. A > B

C. A < B

D. A + 1 < B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q1
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ben and Cal are next to each other"?

A. B - C = 1
B. |B - C| = 1
C. |B - C| = 2
D. |B - C| <= 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q2
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. With five lockers, the clue "Eli is not at either end" means E is 2, 3 or 4.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q3
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The clue "Dee is directly left of Ava" means the same as D < A.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q4
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Dee is in locker 2, and exactly one locker lies between Cal and Dee. Which locker is Cal in?

Locker: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q5
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "At least two lockers lie between Ava and Eli"?

A. |A - E| >= 3

B. A - E >= 3

C. |A - E| = 3

D. |A - E| >= 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q1
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Which condition matches the clue "Ben is in locker 3 unless Cal is"?

A. C = 3

B. B = 3 or C = 3

C. B is not 3 and C is not 3

D. B = 3 and C is not 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q2
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Six friends, Ava, Ben, Cal, Dee, Eli and Fay, each use one of six lockers numbered 1 to 6 from left to right. "Left of" means a lower number. Clues: (1) Ava is somewhere left of Ben. (2) Exactly two lockers lie between Cal and Dee. (3) Fay is somewhere right of Cal. (4) Dee is in an odd-numbered locker. (5) Cal is somewhere left of Ava. (6) Ben is not in locker 6. Put the friends in locker order, from locker 1 to locker 6.

Put these in order:

A Ava
B Ben
C Cal
D Dee
E Eli
F Fay

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q3
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The two conditions A < B and B < A + 2 together say that Ben is directly right of Ava.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q4
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Seven lockers are numbered 1 to 7 from left to right. Give lockers for Ava and Ben that meet every clue: Ava is somewhere left of Ben; at least two lockers lie between them; Ben's locker is even-numbered.

A = _____

B = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q5
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. If clue 3 is removed, how many arrangements fit the other four clues?

Arrangements: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q1
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. How many arrangements fit both clues "Eli is not at either end" and "Ben and Cal are next to each other" (and no other clues)?

Arrangements: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q2
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. If clue 5 is replaced by "Dee is in locker 2", the puzzle still has exactly one answer, and it is the same answer.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q3
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava and Ben are not both at the ends"?

A. Exactly one of Ava and Ben is in locker 1 or 5

B. Ava and Ben are both in lockers 2, 3 or 4

C. Neither Ava nor Ben is in locker 1 or 5

D. At most one of Ava and Ben is in locker 1 or 5

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q4
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Under the clue "Ben is in locker 3 unless Cal is", an arrangement with Ava in locker 3 is allowed.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q5
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Set 35: Must, could, cannot

Local outcome LOGIC.CONSTRAINT.03: Decide what must be true, what could be true and what cannot be true in a constrained arrangement

Puzzle set · genre: scheduling puzzles · MP1, MP3

- MAIN: Decide what must be true, what could be true and what cannot be true in a constrained arrangement
- BACKWARD 1: Strategy: list every case, then read must, could, cannot
- FORWARD 1: ADVANCED APPLICATION: Must, could, cannot under an added condition
- FORWARD 2: EXPERT CHALLENGE: Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement MUST be true?

A. Kites is on Friday.

B. Fossils is on Monday.

C. Gears is on Monday or Tuesday.

D. Honey is on Friday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q1
Logic, Set 35: Must, could, cannot · MAIN · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement CANNOT be true?

A. Jazz is on Tuesday.

B. Kites is on Wednesday.

C. Fossils is on Friday.

D. Honey is on Wednesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q2
Logic, Set 35: Must, could, cannot · MAIN · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Fossils could be on Friday.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q3
Logic, Set 35: Must, could, cannot · MAIN · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many complete schedules meet all four conditions?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q4
Logic, Set 35: Must, could, cannot · MAIN · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Wednesday. Using Monday = 1, Tuesday = 2, Wednesday = 3, Thursday = 4 and Friday = 5, give a day number Kites could be on.

Kites day = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q5
Logic, Set 35: Must, could, cannot · MAIN · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. How many schedules meet both conditions?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q1
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then Q must be on day 1.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q2
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then R must be on day 3.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q3
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

In a scheduling puzzle, what does it mean to say a statement CANNOT be true?

A. It is not mentioned in any of the conditions.
B. It is false in every schedule that meets the conditions.
C. It is false in at least one schedule that meets the conditions.
D. It is true in exactly one schedule that meets the conditions.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q4
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Which statement COULD be true?

A. P is on day 1.
B. R is on day 1.
C. P is on day 2.
D. Q is on day 3.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q5
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Monday, which statement MUST be true?

A. Fossils is on Wednesday.

B. Jazz is on Thursday.

C. Honey is on Friday.

D. Gears is on Tuesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q1
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Friday, how many schedules are possible?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q2
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Jazz is on Wednesday, then Honey must be on Friday.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q3
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which ONE added condition would leave exactly one possible schedule?

A. Fossils is on Wednesday.

B. Kites is on Thursday.

C. Fossils is on Friday.

D. Honey is on Wednesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q4
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Friday. Using Monday = 1 to Friday = 5, give the day numbers of Gears and Jazz in one schedule that meets every condition.

Gears day = _____

Jazz day = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q5

Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many of the five talks could be on Friday?

Talks: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q1
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose condition 3 is replaced by "Fossils is on Monday." Which statement must be true?

A. Kites is on Friday.

B. Honey is on Friday.

C. Jazz is on Wednesday or Thursday.

D. Gears is on Tuesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q2
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Removing condition 4 adds exactly two new schedules.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q3
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Add a fifth condition: no two of Gears, Honey and Jazz are on consecutive days. How many schedules remain?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q4
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Gears is on Monday, then Fossils and Kites could be on Tuesday and Thursday.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q5
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Set 36: Search every case

Local outcome LOGIC.SEARCH.01: Organize a complete case analysis and count or find every solution

Puzzle set · genre: complete search · MP1, MP8

MAIN: Organize a complete case analysis and count or find every solution
BACKWARD 1: Strategy: make an organized list
FORWARD 1: ADVANCED APPLICATION: Counting with extra conditions
FORWARD 2: EXPERT CHALLENGE: Searches with many cases

How many three-digit whole numbers (100 to 999) have digits that add up to 5?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q1
Logic, Set 36: Search every case · MAIN · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 22?

Times: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q2
Logic, Set 36: Search every case · MAIN · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Find the two-digit whole number that equals 3 times the sum of its digits.

Number = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q3
Logic, Set 36: Search every case · MAIN · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can you make 30 cents using nickels (5 cents), dimes (10 cents) and quarters (25 cents)? You may use any number of each coin, including none; the order of the coins does not matter.

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q4
Logic, Set 36: Search every case · MAIN · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Exactly 6 of the 24 orderings of the letters A, B, C and D have A before B and also C before D.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q5
Logic, Set 36: Search every case · MAIN · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many two-digit numbers have digits that add up to 6?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q1
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can you make 20 cents using only nickels and dimes? The order of the coins does not matter.

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q2
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

There are exactly 4 two-digit numbers whose digits multiply to 12.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q3
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Ana lists the ways to make 15 cents from nickels and dimes: "3 nickels; 1 dime and 1 nickel." Is her list complete? The order of the coins does not matter.

A. No: a nickel then a dime is a third way

B. No: 2 dimes and no nickels is missing

C. No: 1 dime alone is missing

D. Yes: she used 0 dimes and 1 dime, and 2 dimes is too much

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q4
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

There are 5 two-digit numbers whose digits add up to 3.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q5
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many three-digit numbers have three different digits that add up to 6?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q1
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can 10 be written as a sum of 1s and 2s if the order matters? (For example, 3 can be written 3 ways: 1 + 1 + 1, 1 + 2 and 2 + 1.)

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q2
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Find a three-digit number whose three digits are all odd, all different, and add up to 15.

Number = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q3
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

There are more ways to make 50 cents from nickels and dimes than from dimes and quarters. The order of the coins does not matter.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q4
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many rectangles of any size (including squares) can be traced along the lines of a grid that is 2 squares tall and 3 squares wide?

A. 15
B. 18
C. 17
D. 6

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q5
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 6?

Times: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q1
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can you make 1 dollar (100 cents) using nickels, dimes and quarters? The order of the coins does not matter.

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q2
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many positive whole numbers less than 1000 have digits that add up to 4?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q3
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Every whole number of cents from 8 to 30 can be paid exactly using only 3-cent and 5-cent stamps.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q4
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Give a whole number of cents from 1 to 10 that CANNOT be paid exactly using only 3-cent and 5-cent stamps.

Amount = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q5
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8